Globally Complete, Locally Grounded: SHAP-Guided Concept Discovery for Neural Networks

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Abstract

Deep neural networks often achieve outstanding performance but lack transparency. Concept-based explanations provide human-friendly insights by discovering vectors (or "concepts") that represent high-level factors in the latent space. A recent framework introduced the notion of *completeness*, quantifying how sufficient a set of concepts is for reconstructing the model's decisions. However, purely completeness-driven approaches may overlook important local phenomena.

In this project, we integrate local attributions from SHAP into concept discovery. We first identify top-influential patches (based on SHAP values) and cluster them by unsupervised learning to obtain *cluster centers*. Then, we introduce a *SHAP-cluster loss* that encourages each concept vector to align with at least one of these centers, ensuring local coherence.

By combining completeness regularization with this SHAP-based clustering, our pipeline yields concepts that are globally sufficient (i.e., maintain high completeness) and locally grounded (anchored to SHAP-driven segments). We demonstrate this approach on IMDB sentiment analysis and on an image dataset, showing that it preserves accuracy while improving interpretability. Overall, this method bridges local and global explanation strategies, producing more coherent and complete concept-based explanations.

1. Introduction

Deep neural networks (DNNs) have achieved state-of-the-art performance in a variety of domains, from image classification to text sentiment analysis. However, their "black box" nature limits their acceptance in critical applications where interpretability is essential. Traditionally, methods such as feature-based attribution (e.g.SHAP [1],LIME [2], and Grad-CAM [3]) provide local explanations by assigning importance scores to individual input features. While these methods are valuable for understanding single-instance predictions, they often fail to capture the global reasoning of the model.

Humans, in contrast, tend to explain decisions using high-level concepts—grouping similar examples and reasoning in terms of abstract ideas. In recent work, researchers have pursued concept-based explanations where latent directions in the activation space are interpreted as concepts (e.g., TCAV [4], ACE [5], and ConceptSHAP [6]). . Such methods assume that the concepts lie in a low-dimensional subspace of an intermediate DNN layer and that a set of these concepts can be sufficient to reconstruct the model's predictions. This property is formalized via a *completeness* score, which measures how well the model's output can be approximated using only the discovered concept scores.

However, our investigations have revealed that the concepts extracted by existing methods may sometimes be difficult for humans to interpret. In some cases, the discovered concepts diverge from the training objectives of the network, capturing features that are not semantically aligned with the task at hand. To mitigate this, we propose an augmentation to the concept discovery pipeline by incorporating local explanations from SHAP. Specifically, we extract high-level embeddings from the most influential tokens (or patches) via SHAP and use clustering techniques (e.g., KMeans) to identify representative *cluster centers* in the latent space. We then introduce a SHAP-cluster loss that penalizes the distance between each cluster center and its closest concept vector. This regularizer encourages the learned concepts not only to be complete in explaining the model's decisions but also to remain aligned with the salient, important high level features.

Thus, our work extends previous concept-based explanation methods by integrating local attribution signals into the global completeness framework. In doing so, we aim to provide explanations that are both faithful to the model and interpretable to human users, bridging the gap between local and global interpretability.

We unify these lines by:

- Using SHAP to identify influential segments (sentences

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in text, patches in images).

- Clustering these top segments to produce "cluster centers."
- Adding a *SHAP cluster loss* in the concept learning objective, ensuring discovered concepts align with these meaningful clusters.

Our pipeline is especially relevant for tasks like IMDB sentiment classification . We show how to adapt the completeness-based objective to incorporate an additional regularizer that enforces closeness between concept vectors and the SHAP-based cluster centers. This yields concept vectors that not only preserve completeness but also match salient patterns discovered via local attributions.

2. Problem Definition

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a dataset where each input $x_i \in \mathcal{X}$ (e.g., an image or a text) has an associated label $y_i \in \mathcal{Y}$. We assume that a pre-trained deep neural network (DNN) $f : \mathcal{X} \to \mathcal{Y}$ can be decomposed into two functions:

$$f(x) = h\big(\phi(x)\big),$$

where $\phi(x) \in \mathbb{R}^d$ denotes an intermediate representation (or feature vector) extracted from a chosen layer, and $h(\cdot)$ maps these features to the final prediction.

Furthermore, we assume that each input x_i can be partitioned into T parts (e.g., image patches or text segments), i.e.,

$$x_i = \{x_{i,1}, x_{i,2}, \dots, x_{i,T}\},$$

so that the overall representation $\phi(x_i)$ is an aggregation of the part-wise features $\{\phi(x_{i,t})\}_{t=1}^T$.

Our objective is to discover a set of m concept vectors $\{c_1, c_2, \dots, c_m\} \subset \mathbb{R}^d$ such that the *concept score*

$$v_c(x_i) = \Big(\langle \phi(x_i), c_1 \rangle,\ \langle \phi(x_i), c_2 \rangle,\ \dots,\ \langle \phi(x_i), c_m \rangle\Big)$$

serves as a sufficient statistic for predicting y_i. In other words, there exists a mapping $g : \mathbb{R}^m \to \mathbb{R}^d$ satisfying

$$h\big(g(v_c(x_i))\big) \approx f(x_i), \quad \forall\, x_i \in \mathcal{D}.$$

Thus, we aim to maximize the log-likelihood (or accuracy) of the model's predictions given only the concept scores while regularizing the learned concepts to ensure both completeness and interpretability:

$$\max_{c_1,\dots,c_m,\, g} \ \log P\big(y \mid g(v_c(x))\big) + R_{\text{reg}},$$

where the overall regularization term R_{reg} is defined as the combination of two components:

Regularization Terms

1. Proximity Regularization: We require that for each concept vector c_k, the top-K nearest neighbor patches (or segments) in the training set are close to c_k. Let T_{c_k} denote the set of the top-K nearest neighbors (according to a similarity measure such as the dot product) of c_k. The proximity regularization term is defined as

$$\begin{aligned} R_{\text{prox}} = \lambda_1 \left(\frac{1}{m} \sum_{k=1}^{m} \frac{1}{K} \sum_{x \in T_{c_k}} \langle \phi(x), c_k \rangle \right) \\ - \lambda_2 \left(\frac{1}{m(m-1)} \sum_{j \neq k} \langle c_j, c_k \rangle \right). \end{aligned}$$

Second term encourages each concept to be strongly associated with a coherent set of input patches while keeping different concepts distinct.Where λ_1 and λ_2 are hyperparameters.

2. SHAP-Cluster Regularization: In addition, we propose a novel regularizer that leverages SHAP-based analysis to capture high-level semantic information from the data. We first compute sentence-level SHAP values for a subset of training samples and then cluster the corresponding high-level embeddings (obtained from an intermediate layer) via K-means clustering. Let $\{\mu_1, \mu_2, \dots, \mu_N\}$ denote the resulting N cluster centers. The SHAP-cluster loss is defined as:

$$R_{\text{shap}} = \lambda_{\text{shap}} \cdot \frac{1}{N} \sum_{i=1}^{N} \min_{j \in \{1,\dots,m\}} \|c_j - \mu_i\|_2,$$

where λ_{shap} is the regularization coefficient. This loss term forces at least one concept vector to be close to each cluster center, thus promoting that the discovered concepts align well with the high-level semantic patterns derived from the SHAP analysis.

Therefore, the complete objective function is given by:

$$\max_{c_1,\dots,c_m,\, g} \ \log P\big(y \mid g(v_c(x))\big) + R_{\text{prox}} - R_{\text{shap}},$$

where the negative sign before R_{shap} indicates that we aim to minimize the distance between the concept vectors and the cluster centers.

Notation Summary:
$\mathcal{D}$ – dataset;
x_i – input sample; y_i – label;
$\phi(x)$ – intermediate activation; $h(\cdot)$ – final prediction function;
c_i – concept vectors; $v_c(x)$ – concept score; g – mapping function;
T_{c_k} – top-K nearest neighbors for concept c_k; μ_i – ith cluster center from SHAP analysis;
$\lambda_1, \lambda_2, \lambda_{\text{shap}}$ – regularization coefficients.

3. Related Work

Interpretable explanations for DNNs have been developed from several different perspectives. Broadly, these methods can be classified as follows:

Feature-Based Explanation Methods. Techniques such as LIME [2] and SHAP [1] assign importance scores to input features. These methods provide local explanations but do not capture the global reasoning process of the network.

Sample-Based Explanation Methods. Other approaches explain predictions by finding similar examples in the training set [7]. While intuitive, these methods lack a formal global framework.

Counterfactual Explanation Methods. Counterfactual methods address the question of "what to change" in the input to alter the model's output [8]. Although they offer actionable insights, they often do not reveal the underlying factors driving the prediction.

Concept-Based Explanations. Concept-Based Explanations. More recent methods seek to interpret DNNs using high-level concepts. For instance, TCAV [4] uses human-annotated concepts to measure directional sensitivities, and ACE [5] automatically discovers concepts via clustering image patches. Other approaches apply unsupervised dimensionality reduction methods such as PCA [9] to obtain latent factors, though these do not necessarily align with human semantics. Furthermore, Locatello et al. [10] emphasize that unsupervised discovery of semantically meaningful latent dimensions requires inductive biases.

In contrast, our approach leverages the completeness framework and enhances it by integrating a SHAP-based clustering regularizer. This new regularizer forces the concept vectors to align with high-SHAP clusters—thereby ensuring that the discovered concepts are both globally sufficient to explain model predictions and locally coherent with influential input segments.

4. Methodology

Our goal is to derive concept-based explanations that are both *complete* (i.e., sufficient to explain the model's predictions) and *interpretable* by humans. The pipeline consists of four main stages: (1) Fine-tuning BERT for sentiment classification and extracting intermediate activations; (2) Computing SHAP values on sentence-level inputs and clustering the high-level embeddings to obtain semantic cluster centers; (3) Training a ConceptNet model with a composite loss that includes both proximity and SHAP-cluster regularizers; and (4) Interpreting the learned concepts via nearest neighbor analysis.

4.1. BERT Fine-Tuning and Activation Extraction

We begin with a pre-trained BERT model which is fine-tuned on the IMDB sentiment classification task. Let

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n},$$

where each review x_i is tokenized and its corresponding label $y_i \in \{0, 1\}$ indicates the sentiment. The fine-tuned model is defined as

$$f(x) = h\big(\phi(x)\big),$$

with $\phi(x) \in \mathbb{R}^d$ representing the intermediate embedding extracted from a chosen BERT layer, and $h(\cdot)$ being the classification head. The training objective is the standard cross-entropy loss:

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{n}\sum_{i=1}^{n} \log P\big(y_i \mid x_i\big).$$

After training, we extract the activations from the penultimate layer (using a forward hook) as a representation of each input.

4.2. SHAP Analysis and Cluster Center Extraction

To capture high-level semantic patterns, we compute SHAP values for a subset of training samples at the sentence level. For a given input sentence x, the SHAP value for feature (token) i is defined as

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!}\Big(f(S \cup \{i\}) - f(S)\Big),$$

where F is the set of tokens in x. In our implementation, we aggregate the absolute SHAP contributions for the positive class to compute an importance score:

$$I(x) = \sum_{i=1}^{|x|} \big|\phi_i^{(1)}\big|.$$

We then rank the sentences by $I(x)$ and select the top-k sentences. Their corresponding BERT embeddings (from the activation extraction step) are clustered using K-means. Let

$$\{\mu_1, \mu_2, \dots, \mu_N\}$$

be the N cluster centers, which capture the dominant high-level semantics derived from the SHAP analysis. These centers will later be used in the SHAP-cluster regularization term.

4.3. ConceptNet Training

Our goal is to learn a set of m concept vectors $\{c_1, c_2, \dots, c_m\} \subset \mathbb{R}^d$ such that the *concept score*

$$v_c(x) = \big(\langle \phi(x), c_1 \rangle,\ \langle \phi(x), c_2 \rangle,\ \dots,\ \langle \phi(x), c_m \rangle\big)$$

is a sufficient statistic for predicting y. That is, there exists a mapping $g : \mathbb{R}^m \to \mathbb{R}^d$ such that

$$h\big(g(v_c(x))\big) \approx f(x), \quad \forall\, x \in \mathcal{D}.$$

Our overall objective is to maximize the log-likelihood of the predictions given the concept scores while enforcing two key regularizations:

$$\max_{c_1,\dots,c_m,\, g} \log P\big(y \mid g(v_c(x))\big) + R_{\text{reg}},$$

with the composite regularizer defined as

$$R_{\text{reg}} = R_{\text{prox}} - R_{\text{shap}}.$$

Proximity Regularization

This term encourages each concept vector to be close to its top-K nearest neighbor patches (or segments) in the training data, while maintaining diversity among different concepts. Formally, for each concept c_k, let T_{c_k} denote its set of top-K nearest neighbors (using, e.g., the dot product as a similarity measure). Then, the proximity regularizer is defined as:

$$\begin{aligned} R_{\text{prox}} = \lambda_1 \left(\frac{1}{m} \sum_{k=1}^{m} \frac{1}{K} \sum_{x \in T_{c_k}} \langle \phi(x), c_k \rangle \right) \\ - \lambda_2 \left(\frac{1}{m(m-1)} \sum_{j \neq k} \langle c_j, c_k \rangle \right), \end{aligned}$$

where λ_1 and λ_2 are hyperparameters that balance the two terms. The first term encourages high similarity between a concept and its neighbors, whereas the second term discourages high similarity among different concept vectors.

SHAP-Cluster Regularization

To integrate high-level semantic information, we introduce a SHAP-cluster regularizer that aligns the learned concept vectors with cluster centers derived from SHAP analysis. Let the cluster centers obtained from K-means clustering be $\{\mu_1, \dots, \mu_N\}$. The SHAP-cluster loss is defined as:

$$R_{\text{shap}} = \lambda_{\text{shap}} \cdot \frac{1}{N} \sum_{i=1}^{N} \min_{j \in \{1,\dots,m\}} \|c_j - \mu_i\|_2,$$

where λ_{shap} is a regularization coefficient. This term minimizes the average distance between each cluster center and its closest concept vector, ensuring that the discovered concepts capture the salient semantic patterns identified by SHAP.

Overall Objective

Thus, the final objective function for ConceptNet training is:

$$\max_{c_1,\dots,c_m,\, g} \log P\big(y \mid g(v_c(x))\big) + R_{\text{prox}} - R_{\text{shap}},$$

or equivalently, we minimize the loss function

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_1 R_{\text{prox}} - \lambda_{\text{shap}} R_{\text{shap}},$$

where the prediction loss is given by

$$\mathcal{L}_{\text{pred}} = -\frac{1}{n} \sum_{i=1}^{n} \log P\Big(y_i \mid h\big(g(v_c(x_i))\big)\Big).$$

4.4. Algorithmic Implementation

The overall process is implemented as follows:

1. **BERT Fine-Tuning and Activation Extraction:** A pre-trained BERT model is fine-tuned on the IMDB dataset. We then extract intermediate activations $\phi(x)$ from the penultimate layer using a forward hook.
2. **SHAP Analysis and Clustering:** We compute sentence-level SHAP values for a subset of the training data using a SHAP explainer. For each sentence, we aggregate the absolute SHAP values (typically for the positive class) to obtain an importance score:

   $$I(x) = \sum_{i=1}^{|x|} \big|\phi_i^{(1)}\big|.$$

   The top-k sentences (based on $I(x)$) are selected, and their corresponding embeddings are clustered using K-means. The resulting cluster centers $\{\mu_1, \dots, \mu_N\}$ serve as proxies for high-level semantic concepts.
3. **ConceptNet Training:** With the BERT embeddings and the derived cluster centers, we train a ConceptNet model to learn a set of concept vectors $\{c_1, \dots, c_m\}$. The ConceptNet loss function comprises (i) a prediction loss based on cross-entropy, (ii) a proximity regularization term R_{prox}, and (iii) a SHAP-cluster regularization term R_{shap}, as described above.
4. **Interpretation and Analysis:** To evaluate the semantic coherence of the discovered concept vectors, we employ a nearest neighbor analysis that operates as follows:
   (a) **Concept Loading:** The learned concept vectors are stored in a NumPy file (e.g., `conceptSHAP/concepts.npy`). These vectors are loaded and transposed to obtain a tensor of size $(m \times d)$, where m is the number of concepts and d is the embedding dimension.
   (b) **Embedding Preparation:** The training embeddings (obtained from BERT) are converted into a torch tensor. Each embedding represents the activation of a training example from an intermediate layer.
   (c) **Nearest Neighbor Search:** For each concept vector c_i, we compute the Euclidean distance between c_i and all training embeddings $\phi(x)$:

   $$d(c_i, \phi(x)) = \|\phi(x) - c_i\|_2.$$

We then retrieve the top 150 nearest neighbors (i.e., the training examples with the smallest distances) for each concept.

(d) **Word Extraction and Frequency Analysis:** For each of the top-150 nearest neighbors, we extract the words from the corresponding input sentence. After filtering out common English stop words and punctuation, we count the frequency of each remaining word.

(e) **Concept Characterization:** Finally, we report the 25 most common words for each concept as a means of interpreting the semantic meaning of that concept.

This process is implemented in our `concept_analysis` function.The resulting word distributions help us assess whether the discovered concepts align with coherent semantic patterns in the data.

4.5. Discussion and Limitations

Although our method successfully integrates local (proximity-based) and global (SHAP-cluster) regularizations to discover semantically meaningful and complete concepts, several challenges remain:

- **Hyperparameter Sensitivity:** The performance of our method depends on the appropriate selection of λ_1, λ_2, λ_{shap}, the number of neighbors K, and the number of clusters N. Extensive tuning may be required for different datasets.
- **Computational Overhead:** The computation of SHAP values and the subsequent K-means clustering add significant computational cost, which may become a bottleneck for large datasets.
- **Multi-Class Considerations:** While our framework is designed for binary classification, extending the method to multi-class settings involves additional complexities (e.g., per-class concept importance scoring).

Notation Summary:
$\mathcal{D}$ – Dataset; x_i – Input sample; y_i – Label; $\phi(x)$ – Intermediate activation; $h(\cdot)$ – Prediction head; c_i – Concept vectors; $v_c(x)$ – Concept score; g – Mapping function; T_{c_k} – Top-K nearest neighbors for concept c_k; μ_i – Cluster centers from SHAP analysis; $\lambda_1, \lambda_2, \lambda_{\text{shap}}$ – Regularization coefficients.

In summary, our methodology integrates traditional BERT fine-tuning with a novel regularization framework that combines local proximity and high-level semantic alignment (via SHAP clustering) to produce concept-based explanations that are both complete and interpretable.

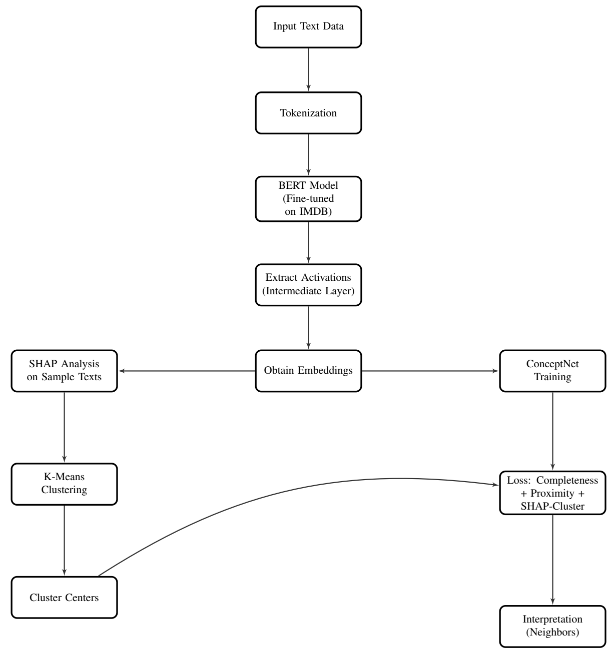

Figure 1. Flowchart of the BERT + ConceptNet pipeline with SHAP-cluster integration.

5. Evaluation

To evaluate our method, we performed experiments on the IMDB movie review dataset using a pre-trained BERT model and the ConceptNet framework described in Section 4. Below, we summarize our main findings and the evaluation procedure:

5.1. Training and Logging

TensorBoard Logs. During training, we tracked several metrics using TensorBoard, shown in Figure 2 and Figure 3:

- **L1 (Proximity) Term:** Gradually increases as concept vectors move closer to their top-K nearest neighbors.
- **L2 (Orthogonality) Term:** Decreases (and becomes negative) as concepts become more distinct from each other.
- **Sum_loss:** The overall training objective, including the cross-entropy loss and regularization terms.
- **Concept Completeness:** Tracks the completeness score of the discovered concepts (as discussed in Section 5.2).

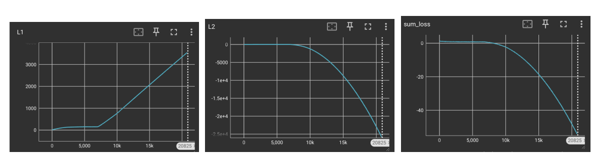

Figure 2. TensorBoard loss plots: L1 (proximity), L2 (orthogonality), and sum of all losses during training.

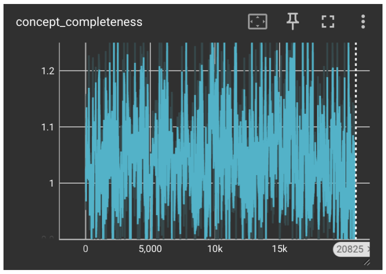

Figure 3. Concept completeness score over training steps.

5.2. Completeness Score

Recall that we define a *completeness* metric to quantify how sufficient the learned concepts are for predicting the label. Concretely, for each mini-batch we compute:

$$\text{Completeness}(y_{\text{pred}}) = \frac{\text{new\_correct} - \frac{1}{m}}{\text{orig\_correct} - \frac{1}{m}}, \qquad (1)$$

where

- orig_correct is the number of correct predictions by the original BERT model on that batch,
- new_correct is the number of correct predictions using only the concept-based projection,
- m is the number of discovered concepts, i.e., `n_concepts`.

This ratio indicates whether the concept-based model approaches the performance of the original network. Values close to 1.0 suggest that the concepts are *sufficient* to recover the underlying predictions.

5.3. Discovered Concepts and Nearest Neighbors

After training, we analyzed each discovered concept by retrieving its top 150 nearest neighbors in the dataset. We used the following snippet to examine the frequent words in these neighbors:

Table 1. Most frequent words associated with each discovered concept.

Concept	Top Words (word, frequency)
Concept 1	dull (27), really (24), stupid (24), movie (22), pointless (19), plot (12), just (11), />br (9), it. (8), thing (8), annoying (8), terrible (7), calculated (6), imitation (6), disguised (6), poorly (6), acted (6), clearly (6), biased (6), funny (6), got (6), idea (6), weak (6), unfortunately (6), it's (6)
Concept 2	dull (23), just (22), really (22), stupid (19), acting (13), horrible (13), movie (13), poorly (11), bad (10), funny (9), absolutely (9), boring (9), ridiculous (8), plot (8), all. (7), story (7), movie. (7), />br (7), simply (6), lacks (6), predictable (6), terrible (6), big (6), disappointment (6), annoying (6)
Concept 3	story (9), good (8), movie (7), better (7), just (6), pretty (6), interesting (5), acting (5), performances (5), film (5), great (4), real (4), quite (4), />br (4), truly (3), character (3), script (3), way (3), hilarious (3), thing (3), act (2), musical (2), strange (2), make (2), world (2)
Concept 4	love (26), wonderful (21), real (18), story (15), really (13), life (13), truly (11), movie (10), entertaining (9), wild (7), good (7), leaves (7), strange (6), weird (6), character (6), moved (5), brave (5), little (5), boy (5), exceptional (5), beautiful (5), gives (5), man (5), book (5), feeling (5)
Concept 5	love (24), wonderful (22), real (18), story (16), really (13), life (13), truly (12), movie (10), entertaining (9), wild (7), good (7), strange (6), weird (6), character (6), man (6), leaves (6), great (6), exceptional (5), moved (5), brave (5), little (5), boy (5), gives (5), beautiful (5), way (5)

From this analysis, we observed that:

- **Concepts 1 & 2 (Negative Sentiment):** Their top words included *"dull"*, *"stupid"*, *"pointless"*, *"bad"*, etc.
- **Concepts 3, 4, 5 (Positive Sentiment):** Their top words included *"love"*, *"wonderful"*, *"real"*, *"great"*, etc.

Hence, the learned concept vectors indeed capture high-level sentiment patterns.

5.4. Summary of Findings

Overall, the training logs (see L1, L2, completeness, and sum_loss in TensorBoard) show that the proximity term increases to ensure concept patches remain tight, while the orthogonality term decreases to push different concepts apart. The completeness metric stabilizes around a value close to (but slightly below) 1.0, indicating that our concept-based projection retains most of the predictive power of the original BERT model. Furthermore, the semantic analysis of top neighbor patches confirms that each concept captures a coherent theme, ranging from strongly negative to positive sentiment.

6. Conclusions and Future Work

In this work, we proposed a method to discover *completeness-aware* concept vectors from an already fine-tuned BERT model on IMDB sentiment classification. By integrating:

- **Local Proximity** and **Orthogonality** constraints to ensure that concepts are both *coherent* and *distinct*,
- **SHAP-based cluster regularization** to align concepts with high-level semantic groups,
- **Completeness** as a measure of how much the concept-based model recovers the original DNN predictions,

we demonstrated that the discovered concepts are both *interpretable* (as validated by nearest neighbor analysis) and *predictively sufficient* (as shown by our completeness metric).

Highlights.

- We successfully recovered concepts that strongly correlate with negative vs. positive sentiments.
- Our completeness score approached values near 1.0, indicating minimal loss of predictive power.

Future Directions. There remain several interesting avenues to explore:

1. **Multi-task or Multi-domain Settings:** Extending the concept discovery framework to datasets with more complex label structures or multiple tasks. Furthermore, incorporating concept-based interpretability into generative large language models (e.g., GPT) offers a promising avenue to align latent generation dynamics with interpretable, human-centric concept spaces, potentially

enhancing controllability and transparency in text generation.

2. **Joint Concept Learning:** Rather than post-hoc, integrating concept regularization directly into the BERT fine-tuning could yield more robust or disentangled concepts.

Overall, our method provides a promising step toward bridging *concept-based interpretability* with state-of-the-art language models, ensuring that the learned concepts remain both semantically meaningful and highly predictive.

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